

# Goktug Aygun

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Andreas Vesaliusstraat, Leuven / Belgium

## EDUCATION

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### **KU Leuven**, Leuven, Belgium

MEng in *Computer Science, Focusing on Artificial Intelligence*

Sep 2025 -Exp. Jun 2027

Coursework: *Natural Language Processing, Genetic Algorithms and Evolutionary Algorithms, Constraint Solving, Distributed Systems, Declarative Languages, Computer Vision, Reinforcement Learning, Information Retrieval and Search Engines, Modelling of Complex Systems, Data Mining*

### **Sabanci University**, Istanbul, Turkey

BSc in *Computer Science and Engineering*, Minor in *Decision & Behavior*

Sep 2021 - Jun 2025

GPA: 3.78 / 4.00

Ranking: 26 / 319

Coursework: *Data Structures & Algorithms, Theory of Computation, Discrete Math, Computer Architecture, Artificial Intelligence, Machine Learning, Operating Systems, Database Systems, Computer Networks, Software Engineering*

Sakip Sabanci Encouragement Award, 2022 & 2023: Full Tuition Scholarship Awarded for Academic Excellence

## TECHNICAL EXPERIENCE

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### **DidactLabs**, *Software Developer Intern*, Leuven, Belgium

Feb 2026 – Present

- Contributing to the backend development of an AI-integrated education platform using TypeScript
- Assisting with feature implementation, bug fixes, and codebase refactoring
- Collaborating with the engineering team to maintain code quality and support ongoing development efforts

### **OBSS**, *Software Developer Intern*, Istanbul, Turkey

Jun – Sep 2024

- Developed full-stack application using React.js and .NET
- Designed REST APIs with authentication and authorization mechanisms

### **Sabanci University**, *Undergraduate Learning Assistant*, Istanbul, Turkey

Sep 2022 – Jan 2024

- Mentored 2000+ students in Python and C++ across multiple semesters
- Led weekly recitation sessions and office hours to reinforce course concepts

## PROJECTS

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### **Cloud-Deployed Service-Oriented Distributed Ticketing System**

Sep 2025 – Jan 2026

- Designed and deployed a fault-tolerant, service-oriented ticketing platform on Google Cloud
- Implemented retry and timeout mechanisms to integrate reliable and unreliable providers

### **Optimization of the Travelling Salesman Problem Using Genetic Algorithms**

Sep 2025 – Jan 2026

- Implemented a Genetic Algorithm solver for the Travelling Salesman Problem using established evolutionary operators
- Investigated trade-offs between convergence speed and solution optimality across different configurations
- Visualized evolutionary progress to analyze performance trends over generations

### **Analyzing Sabanci University Curricula in Relation to the Sustainable Development Goals**

Nov 2024 – Jun 2025

- Leveraging keyword based classification and LLMs to investigate the alignment between SDGs and course curricula

**Designing and Implementing an LLM-based Course Assistant**, Graduation Project (ENS491) Sep 2024 – Jun 2025

- Creating an LLM-based assistant, aiming to assist students and assistants in Sabanci University
- Adapting LLaMA, creating available and trustworthy AI based assistant using Retrieval Augmented Generation

**Flight Roster Project**, Software Engineering Course (CS308) Feb – Jun 2024

- Built a web application simulating airline operations, architecting the backend and developing RESTful APIs using Node.js while leading a backend team of 7
- Designed and deployed MySQL databases on AWS Cloud Services and leveraged MongoDB for NoSQL solutions, ensuring reliable and efficient data storage and retrieval

**Bug Severity Detector Project**, Machine Learning Course (CS412) Feb – Jun 2024

- Created a classifier utilizing different models with a team of 4
- Aimed to detect the severity of a given bug in range [1,7] based on given textual description

**Investigating Graph Coloring Project**, Algorithms Course (CS301) Feb – Jun 2024

- Collaborated with a team of 2 to investigate graph coloring problem with various heuristic and greedy algorithms
- Implemented and benchmarked heuristic and greedy algorithms for graph coloring

**Study Buddy Mobile Application Project**, Mobile Application Development Course (CS310) Sep 2023 – Jan 2024

- Built a collaborative learning application connecting students enrolled in similar courses, enabling posting and course filtering features
- Designed and developed the Android front-end and Java/Spring-based backend, containerizing services with Docker

**DiSUCord - Message Networking via Channels**, Computer Networks Course (CS408) Sep 2023 – Jan 2024

- Employed TCP sockets and designed server application layer for users to interact with using C# for Graphic User Interfaces besides .NET for necessary infrastructure
- Developed an interactive communication platform for students

**Personal Portfolio Website** Jun 2023 – Jul 2024

- Created a personal website, goktugaygun.com, using HTML, CSS, JS, React and Bootstrap framework

**Analysis of Female MPs in Turkish Assembly**, Introduction to Data Science Course (CS210) Jan – Jun 2023

- Scraped and analyzed data from the Grand Assembly of Turkey to investigate gender inequality, leading a team of 5
- Generated visualizations and reports using Matplotlib and Seaborn to communicate key analytical findings

**Deaths Around the World by Cause Database Application**, Database Systems Course (CS306) Jan – Jun 2023

- Designed and built relational models and database system to store real world data obtained from ourworldindata.org and investigated it using Python with visualization tools

## PROFESSIONAL SKILLS

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- **Programming Languages:** Python, C++, C#, Java, JavaScript, SQL
- **Web & Backend Technologies:** REST APIs, Flask, Spring, .NET, Node.js, React.js
- **Databases:** MySQL, MongoDB, NoSQL, Chroma, Firebase
- **Cloud & DevOps:** Google Cloud, Docker, Git
- **AI & ML:** Langchain, LLMs, RAG, NLP, Pandas, Matplotlib, Selenium, Hugging Face, PyTorch, Scikit-Learn
- **Soft Skills:** Problem Solving, Analytical Thinking, Communication, Team Collaboration, Time Management
- **Languages:** Turkish (Native), English (Advanced, Scored 113 in TOEFL), German (Beginner)

## REFERENCES

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Available upon further request